FH Salzburg
MultiMediaTechnology

Basics Game Development

GameDesign, GUI

Technology
Health
Media

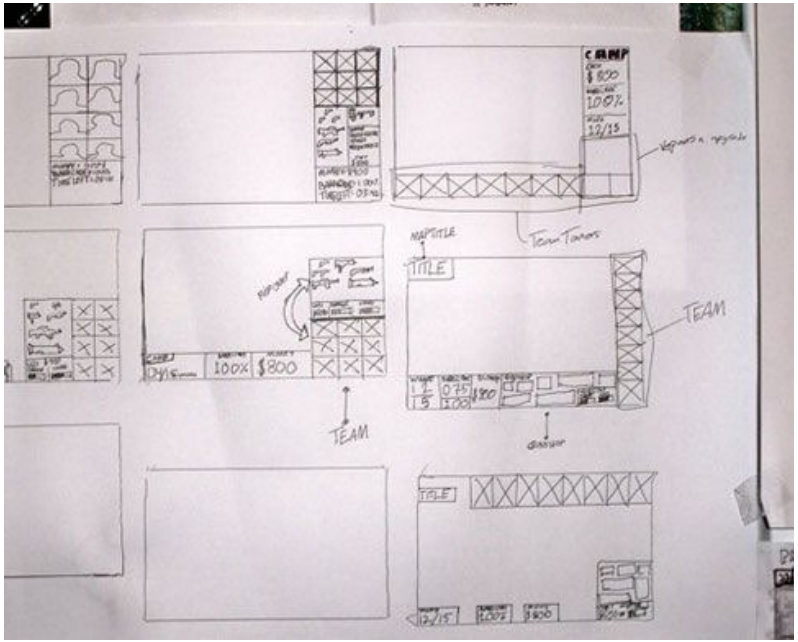
Graphical User Interface (GUI)

Game UI

- Like websites or mobile apps, video games have common UI components that help players navigate and accomplish goals.
- Be very strict with what you add to your UI, **keep it light and simple**. Each additional element adds more visual noise, which will add more cognitive load for the user. Your goal should be for the player to consume more of the game, not wade through a swampy UI.
- Unless your game is story heavy, keep in mind that the **majority will NOT read**.
- <u>Get Inspired!</u>

Design Principles

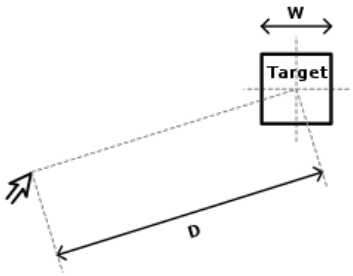
- **Apply design principles from „basics media design“** (1. semester)
- Start with a wireframe and iterate over your UI design. Graphics memory is a rare good.
- Reuse UI-elements and textures as much as possible. Graphics memory is a limited resource.
- Avoid transparencies and stacked textures.
- Stick to a color theme (color.adobe.com)
- Optimize UI-performance



Scale of UI-Components – Fitts‘s Law

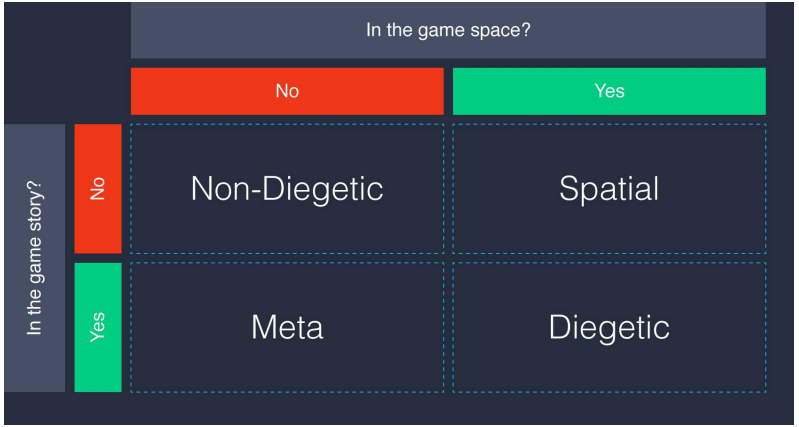
- Fitts's law (often cited as Fitts' law) is a **predictive model of human movement** primarily used in human–computer interaction and ergonomics.
- The theorem states that the **time needed to reach a surface is proportional to the distance from that surface, and to its size**.
- $MT = a + b \cdot ID = a + b \cdot log_2 \left(\frac{2D}{W}\right)$

  MT: average time to complete the movement.
  a and *b*: constants that depend on the choice of input device
  ID: index of difficulty
  D: distance from the starting point to the center of the target
  W: width of the target measured along the axis of motion.



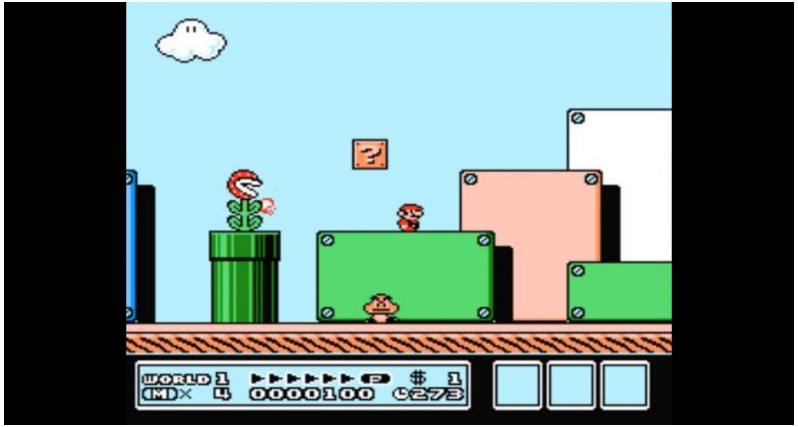
Classes of UI-Components

- GameDesign concepts to adress:
  - **Narrative:** The Story a video game tells.
  - **The Fourth Wall:** imaginary barrier between player and the space in which the game takes place.
- Questions to answer:
  - **Does the component exist in the game story? (Narrative)**
  - **Does the component exist in the game space? (The Fourth Wall)**
- Four classes of ui components: **Non-diegetic, Diegetic, Spatial,** and **Meta**



Non-Diegetic UI Components

- **Non-diegetic UI components reside outside of a game's story and space**. None of the characters in the game, including a player's avatar, are aware that the components exist. The design, placement, and context of non-diegetic components are paramount.
- In fast-paced games, non-diegetic components may interrupt a player's sense of immersion. But in strategy-heavy games, they can provide players with a more nuanced assessment of resources and actions.
- Non-Diegetic components commonly appear in video games as stat meters

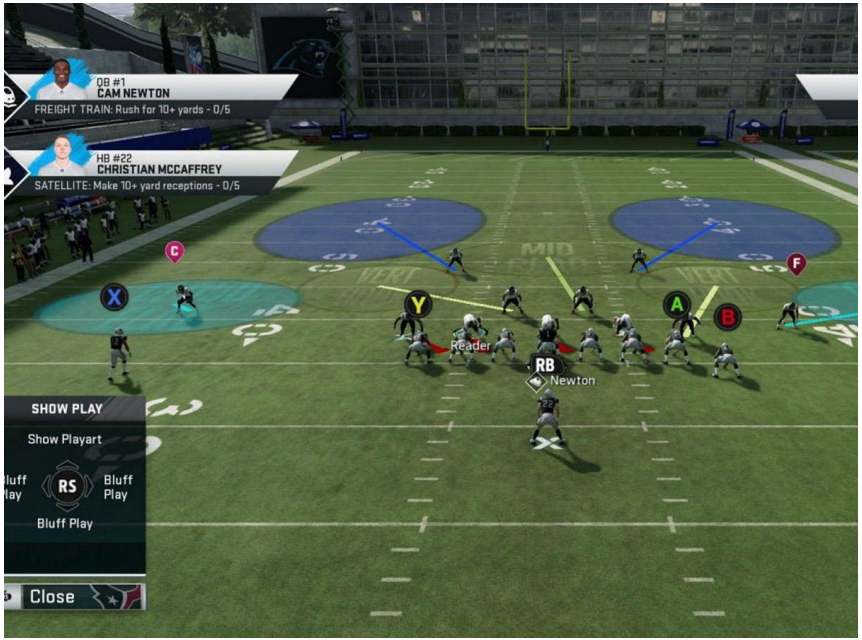


Diegetic UI Components

- Diegetic UI components inhabit both a game's story and space, and characters within the game are aware of the components. Even though they exist within the game story and space, poorly considered diegetic components are still capable of distracting or frustrating players.
- Scale makes diegetic components tricky. For instance, an in-game speedometer that resides on a vehicle's dashboard will likely be too small for players to see clearly. In some games, handheld diegetic components (like maps) can be toggled to a 2-D, full-screen view, making them non-diegetic.

Spatial UI Components

- Spatial UI components are found in a game’s space, but characters within the game don’t see them. Spatial components often work as visual aids, helping players select objects or pointing out important landmarks.
- Text labels are a classic example of spatial UI components. In fantasy and adventure games, players may encounter important objects that are unfamiliar in appearance. Text labels quickly remove ambiguity and keep players immersed in the gaming experience.



Meta UI Components

- Meta UI components exist in a game's story, but they don't reside in the game's space. A player's avatar may or may not be aware of meta components. Traditionally, meta components have been used to signify damage to a player's avatar.
- Meta components can be quite subtle - like a slowly accumulating layer of dirt on the game's 2D plane, but they can also feature prominently in the gaming experience. In action and adventure games, the entire field of view is sometimes shaken, blurred, or discolored to show that a player has taken on damage.

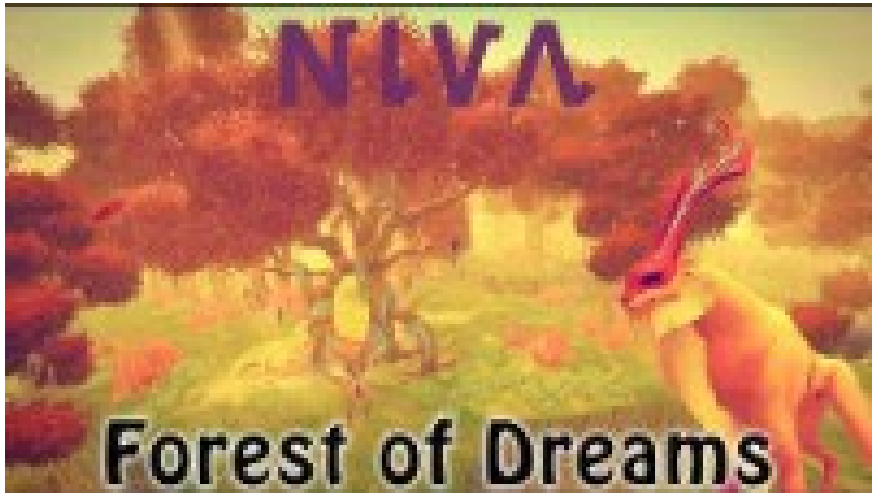
Implementation Workflow

- The workflow differs depending on the library used.
- Both graphical and CSS-like workflows are common.
- TGUI is a popular GUI-Library for SFML (comes with a GUI editor but .NET bindings are outdated)

UI-Component Examples

Consistency in UI Design

- Funny reaction to a button in my master project (~39:00)😂



Localization

- **Localization is the process of customizing an application, webpage, or website for a given culture or locale**.
- Common Procedure:
  - Store all your UI-Strings from the beginning in a structured document format (csv, json, xml, yaml, …).
  - Use tokens for dynamic dialogs.
  - Translate your „String-Storage“ to other languages.
  - At Runtime, you simply need to pick the correct string storage.

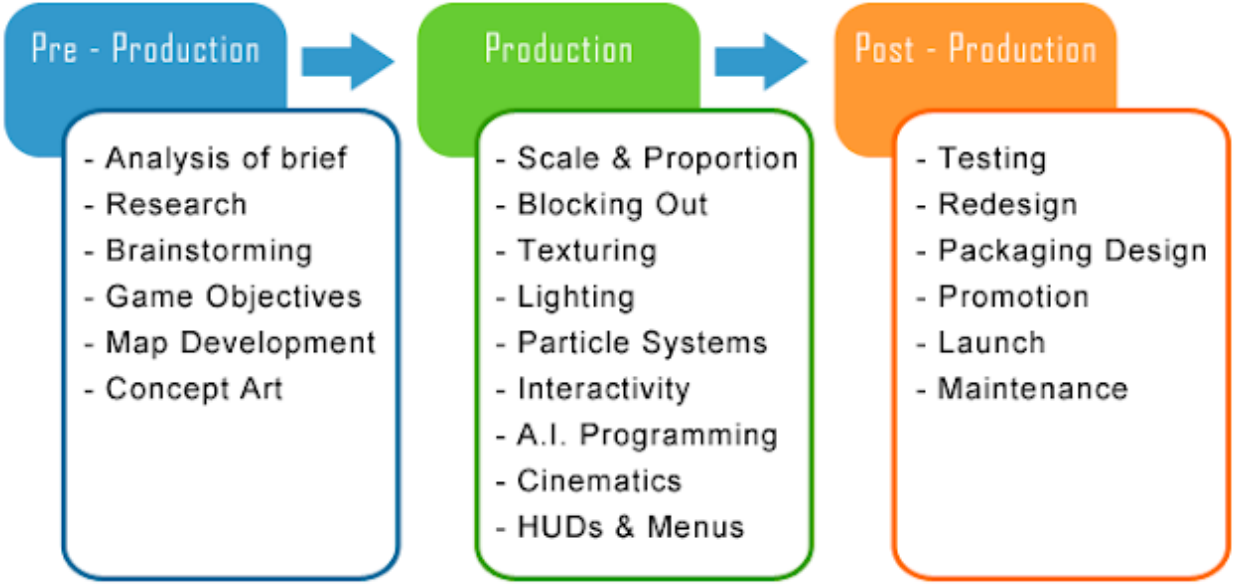
Game Design

What is Game Design?

- Game design can be considered the planning arm of the entire process for making a video game. No video game gets made without a plan, and game design is, more or less, the process of making that plan. The field is somewhat a hybridization of creativity and technical skills that combine into a cohesive, fleshed out idea that people can work with using concrete and actionable tasks. (gamedevacademy.org)

Side note: **Game studies**, also known as **ludology**, is the study of games, the act of playing them, and the players and cultures surrounding them.

Classic Game Design Process



Pre-Production

- **Analysis of Brief/Sector**
  Identify key requirements, analyse market, identify gaps, choose genre, engine, audience, …
- **Research**
  Possible software to use; ideas to develop; competitors on market …
- **Brainstorming**
  Idea generation → difficult and time-consuming task
- **Game Concepts/Objectives:**
  What is the objective of the game? How many levels? Story? Names?...
- **Concept Art:**
  Style of the game, Target-Rendering,
- **Map Development**
  Layout of the game world;

Production

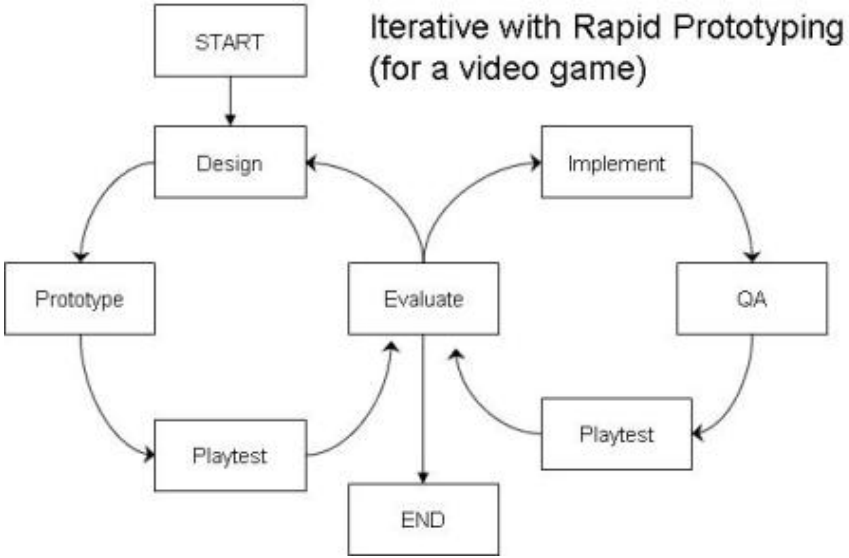
- **Scale and Proportion**
  What are the proportions in the game compared to the character
- **Blocking out the Level**
  Quick layout of main sections and areas of the level. Allows you to get a sense of the environment.
- **Adding Content**
  - Textures,
  - Lighting,
  - Functionality and Interactivity,
  - A.I.,
  - Cinematics,
  - GUI, …

Post-Production

- **Testing**
  Find and fix bugs or glitches
- **Redesign**
  Based on the findings during the testing process various elements may need to be fixed or redesigned.
- **Packing**
  compiling the game folders and files into an executable version of the game
- **Promoting**
  Without good promotion, no matter how good the game is nobody will ever hear about it and therefore will never get the chance to play it.

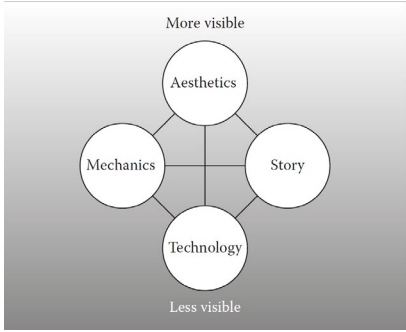
Iterative Game Design Process

- Iterative design is a **play-based design process**. Emphasizing playtesting and prototyping, iterative design is a method in which design decisions are made based on the experience of playing a game while it is in development. (Rules of Play, Game Design Essentials)
- **Rapid Prototyping is a common method in game development.**



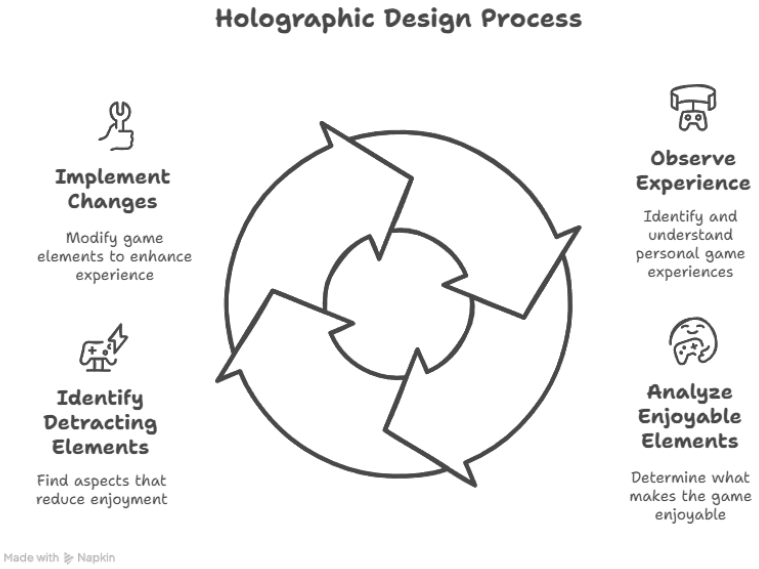
The Four Basic Elements

1. **Mechanics**
   These are the procedures and rules of your game. Mechanics describe the goal of your game, how players can and cannot try to achieve it, and what happens when they try.
2. **Story**
   This is the sequence of events that unfolds in your game. It may be linear and pre-scripted, or it may be branching and emergent.
3. **Aesthetics**
   This is how your game looks, sounds, smells, tastes, and feels. Aesthetics are an incredibly important aspect of game design since they have the most direct relationship to a player's experience.
4. **Technology**
   We are not exclusively referring to "high technology" here, but to any materials and interactions that make your game possible such as paper and pencil, plastic chits, or high-powered lasers.



Holographic Design

- Essentially, the skill you need to develop is the ability to **observe your own experience while thinking about the underlying causes of that experience**. This important **skill is called holographic design**.
  - What elements of the game make the experience enjoyable?
  - What elements of the game detract from the experience?
  - How can I change game elements to improve the experience?

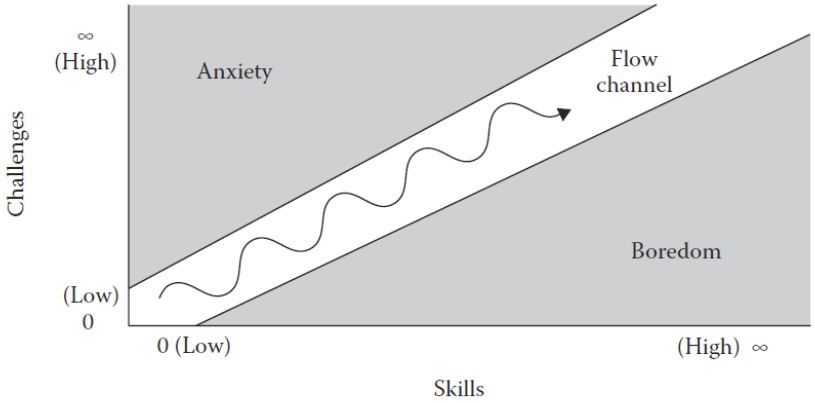


Player Centered Design

- **Everything in Game Design must fit the players intended experience.**
- **Stop thinking about your game and start thinking about your player!**
  - In general, what do they like?
  - What don’t they like? Why?
  - What do they expect to see in a game?
  - If I were in their place, what would I want to see in a game?
  - What would they like or dislike about my game in particular?

Flow

- Flow is sometimes defined as **“a feeling of complete and energized focus in an activity, with a high level of enjoyment and fulfillment.”**
- Flow often appears in tandem with both presence and immersion
  (e.g., “Immersion, engagement and presence” (McMahan, 2003); and “A review on neuropsychophysiological correlates of flow” (Nah et al., 2014))
- 📏 Validated Questionnaires: EGameFlow GAMEFULQUEST



The Art of Game Design – A Book of Lenses

Presence and Immersion

- Immersive games are video games that transport the player into an alternative world where techniques are used to make them feel more like the character they're playing.
- “Immersion is a process of temporarily expanding consciousness into areas of the unconscious—something like hypnosis, but retaining consciousness as one does in lucid dreaming states.” (Handbook of Research on ICTs for Human-Centered Healthcare and Social Care Services)
- Presence, while having similarities to feelings of immersion, is defined as the feeling of existing (at least momentarily) within the game. This exists on a continuum, with complete separation at one end, and strong feelings of physical existence within the game at the other end.
- 📏 Validated Questionnaires: PQ UCLIC

Engagement

- Game engagement refers to the player's' commitment to the gaming activities. Deeply engaged player is fully focused on the gaming activities and is not aware of the things taking place around. **Immersion and flow are central concepts** in many studies **related to the game engagement**.
- Game engagement is a term used to describe **the experience of individual who is fully involved in a mental procedure or physical activity** such as video game playing.
- **Three key components** to increase engagement:
  - **Objective**
  - **Challenge**
  - **Reward**
- 📏 Validated Questionnaires: Game Engagement Questionnaire (GEQ)

Why do loot boxes work?

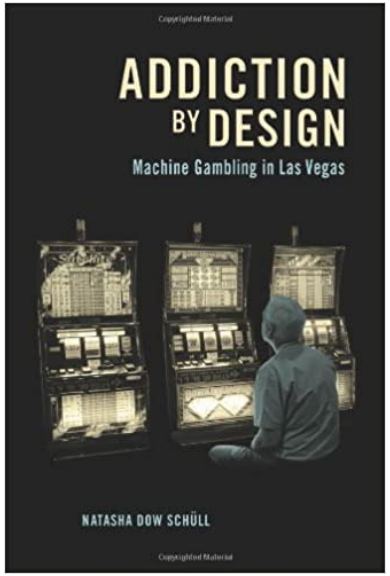
The psychology of loot

- Why do loot boxes provide such a dark compulsion?
  - Psychologists call the principle by which they work on the human mind '**variable rate reinforcement**.'
  - *"The player is basically working for reward by making a series of responses, but the rewards are delivered unpredictably,"* (Dr Luke Clark)
  - *"We know that the **dopamine system**, **which is targeted by drugs** of abuse, is also **very interested in unpredictable rewards**. Dopamine cells are most active when there is maximum uncertainty, and the dopamine system responds more to an uncertain reward than the same reward delivered on a predictable basis."* (Dr Luke Clark)
  - The clever - or insidious - bit is how a loot box is wired into a game, and how it doles out its baubles, keeping a player on the knife-edge between feeling hungry and feeling rewarded.

Addictive Design

- A great game seduces its player into flow state.
- When something is designed to addict people, whether it is a cigarette or a game or a slot machine, a whole range of different mechanisms can be used.
- Some games are more addictive than others simply because they are more rewarding: more interesting, more delightful, or more prosocial.
- **“Addictive Design”** is a commonly applied technique in **“gen-z social media applications”**.
- 📽️ Stanford Seminar – How to Design Addictive Games

2019 the WHO considered video game addiction officially as a mental disorder!



Ethics of Game Design

- Game Designers face the same **moral issues** that artists in any other communications medium face.
- They must struggle with balancing their rights to free expression with the tastes of consumers and be concerned about the **effects their content has on their audience**.
- Legally, games qualify as a form of expression that is protected.
- Three major areas where ethics play a role:
  - Violence
  - Nudity
  - Criminality

Gamification

Wikipedia Definition: **Gamification** is the strategic attempt to enhance systems, services, organizations, and activities in order to create similar experiences to those experienced when playing games in order to motivate and engage users.[1] **This is generally accomplished through the application of game-design elements and game principles** (dynamics and mechanics) in non-game contexts.[2][3] ,,,

Periodic Table of Gamification Elements

Gamified UK

1 Rr Random Rewards									2 Fr Fixed Reward	3 Td Time Dependent
4 Ob On-boarding	5 Si Signposting	6 La Loss Aversion	7 I Investment				8 Pf Progress / Feedback	9 T Theme	10 N Narrative	11 C Curiosity
12 Tp Time Pressure	13 S Scarcity	14 St Strategy	15 F Flow	16 Co Consequences	17 Gt Guilds / Teams	18 Sn Social Network	19 Ss Social Status	20 Sd Social Discovery	21 Sp Social Pressure	22 Cm Competition
23 Ch Challenges	24 Ce Certificates	25 L Learning	26 Q Quests	27 Lp Levels / Progression	28 Bb Boss Battles	29 E Exploration	30 Bc Branching Choices	31 Ee Easter Eggs	32 U Unlockables	33 Ct Creativity Tools
34 Cu Customisation	35 Ap Altruistic Purpose	36 Cg Care Taking	37 A Access	38 Cn Collection	39 Gs Gifting / Sharing	40 Ks Knowledge Share	41 P Points	42 Pr Prizes	43 Le Leaderboards	44 B Badges
		45 Ve Virtual Economy	46 Lo Lottery	47 Ip Innovation Platform	48 V Voting	49 Dt Development Tools	50 A Anonymity	51 Lt Light Touch	52 An Anarchy	

Reward Schedule	General	Socialiser	Achiever	Free Spirit	Philanth-ropist	Player	Disruptor

Game Design Books

- The Art of Game Design: A Book of Lenses ⭐
- Game Design Workshop
- The Ultimate Guide to Video Game Writing and Design
- Level Up! The Guide to Great Video Game Design
- Rules of Play: Game Design Fundamentals
- A Theory of Fun for Game Design
- The Gamification of Learning and Instruction Fieldbook
- Game Feel: A Game Designer’s Guide to Virtual Sensation
- Challenges for Game Designers

We have only to watch young dogs to see that all the essentials of human play are present in their merry gambols. They invite one another to play by a certain ceremoniousness of attitude and gesture. They keep to the rule that you shall not bite, or not bite hard, your brother's ear. They pretend to get terribly angry. And-what is most important-in all these doings they plainly experience tremendous fun and enjoyment. Such rompings of young dogs are only one of the simpler forms of animal play. There are other, much more highly developed forms: regular contests and beautiful performances before an admiring public.

Here we have at once a very important point: even in its simplest forms on the animal level, play is more than a mere physiological phenomenon or a psychological reflex. It goes beyond the confines of purely physical or purely biological activity. It is a significant function-that is to say, there is some sense to it. In play there is something "at play" which transcends the immediate needs of life and imparts meaning to the action. All play means something.

-**Johann Huizinga**, Homo Ludens

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